

CONFIGURE IPV6 ADDRESSING

1. Assign a Static IPv6 Address

Linux Commands

ip a

ifconfig

```

kali@kali: ~
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:1a:e4:4d brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.50/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 86361sec preferred_lft 86361sec
    inet6 2401:4900:8fed:de38:e61f:b06e:aba3:c43e/64 scope global dynamic noprefixroute
        valid_lft 83373sec preferred_lft 83373sec
    inet6 fe80::c677:7fec:f0d2:dfbc/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

kali@kali: ~
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.50 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::c677:7fec:f0d2:dfbc prefixlen 64 scopeid 0x20<link>
    inet6 2401:4900:8fed:de38:e61f:b06e:aba3:c43e prefixlen 64 scopeid 0x0<global>
    ether 00:0c:29:1a:e4:4d txqueuelen 1000 (Ethernet)
    RX packets 199 bytes 59556 (58.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 89 bytes 39976 (39.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

kali@kali: ~

```

2. Configure a Unique Local IPv6 Address (ULA)

Verification Commands

ping6

```

kali@kali: ~
$ ping6 2401:4900:8fed:de38:e61f:b06e:aba3:c43e
PING 2401:4900:8fed:de38:e61f:b06e:aba3:c43e (2401:4900:8fed:de38:e61f:b06e:aba3:c43e) 56 data bytes
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=1 ttl=64 time=0.834 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=2 ttl=64 time=0.852 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=3 ttl=64 time=0.878 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=4 ttl=64 time=0.879 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=5 ttl=64 time=0.851 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=6 ttl=64 time=0.864 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=7 ttl=64 time=0.878 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=8 ttl=64 time=0.854 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=9 ttl=64 time=0.862 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=10 ttl=64 time=0.869 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=11 ttl=64 time=0.868 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=12 ttl=64 time=0.883 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=13 ttl=64 time=0.873 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=14 ttl=64 time=0.838 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=15 ttl=64 time=0.877 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=16 ttl=64 time=0.882 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=17 ttl=64 time=0.869 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=18 ttl=64 time=0.894 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=19 ttl=64 time=0.860 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=20 ttl=64 time=0.836 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=21 ttl=64 time=0.862 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=22 ttl=64 time=0.898 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=23 ttl=64 time=0.864 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=24 ttl=64 time=0.849 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=25 ttl=64 time=0.861 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=26 ttl=64 time=0.827 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=27 ttl=64 time=0.878 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=28 ttl=64 time=0.870 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=29 ttl=64 time=0.860 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=30 ttl=64 time=0.881 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=31 ttl=64 time=0.865 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=32 ttl=64 time=0.863 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=33 ttl=64 time=0.816 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=34 ttl=64 time=0.868 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=35 ttl=64 time=0.873 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=36 ttl=64 time=0.866 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=37 ttl=64 time=0.851 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=38 ttl=64 time=0.865 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=39 ttl=64 time=0.857 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=40 ttl=64 time=0.864 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=41 ttl=64 time=0.817 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=42 ttl=64 time=0.862 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=43 ttl=64 time=0.833 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=44 ttl=64 time=0.827 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=45 ttl=64 time=0.829 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=46 ttl=64 time=0.836 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=47 ttl=64 time=0.813 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=48 ttl=64 time=0.861 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=49 ttl=64 time=0.826 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=50 ttl=64 time=0.860 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=51 ttl=64 time=0.885 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=52 ttl=64 time=0.860 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=53 ttl=64 time=0.885 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=54 ttl=64 time=0.890 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=55 ttl=64 time=0.863 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=56 ttl=64 time=0.891 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=57 ttl=64 time=0.863 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=58 ttl=64 time=0.862 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=59 ttl=64 time=0.862 ms

```

Ping -6

```

kali@kali:~$ cat /dev/null > /home/kali/.ssh_history
kali@kali:~$ ping -6 2401:4900:8fed:de38:e61f:b06e:aba3:c43e
PING 2401:4900:8fed:de38:e61f:b06e:aba3:c43e (2401:4900:8fed:de38:e61f:b06e:aba3:c43e) 56 data bytes
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=1 ttl=64 time=0.872 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=2 ttl=64 time=0.878 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=3 ttl=64 time=0.881 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=4 ttl=64 time=0.887 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=5 ttl=64 time=0.898 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=6 ttl=64 time=0.886 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=7 ttl=64 time=0.879 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=8 ttl=64 time=0.884 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=9 ttl=64 time=0.856 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=10 ttl=64 time=0.859 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=11 ttl=64 time=0.845 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=12 ttl=64 time=0.851 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=13 ttl=64 time=0.864 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=14 ttl=64 time=0.899 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=15 ttl=64 time=0.861 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=16 ttl=64 time=0.845 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=17 ttl=64 time=0.882 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=18 ttl=64 time=0.831 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=19 ttl=64 time=0.861 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=20 ttl=64 time=0.844 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=21 ttl=64 time=0.831 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=22 ttl=64 time=0.831 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=23 ttl=64 time=0.827 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=24 ttl=64 time=0.834 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=25 ttl=64 time=0.881 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=26 ttl=64 time=0.845 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=27 ttl=64 time=0.841 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=28 ttl=64 time=0.847 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=29 ttl=64 time=0.897 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=30 ttl=64 time=0.843 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=31 ttl=64 time=0.893 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=32 ttl=64 time=0.841 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=33 ttl=64 time=0.853 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=34 ttl=64 time=0.849 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=35 ttl=64 time=0.951 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=36 ttl=64 time=0.844 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=37 ttl=64 time=0.888 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=38 ttl=64 time=0.868 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=39 ttl=64 time=0.889 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=40 ttl=64 time=0.844 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=41 ttl=64 time=0.889 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=42 ttl=64 time=0.868 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=43 ttl=64 time=0.863 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=44 ttl=64 time=0.831 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=45 ttl=64 time=0.881 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=46 ttl=64 time=0.847 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=47 ttl=64 time=0.839 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=48 ttl=64 time=0.841 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=49 ttl=64 time=0.876 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=50 ttl=64 time=0.861 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=51 ttl=64 time=0.871 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=52 ttl=64 time=0.829 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=53 ttl=64 time=0.886 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=54 ttl=64 time=0.847 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=55 ttl=64 time=0.887 ms
64 bytes from 2401:4900:8fed:de38:e61f:b06e:aba3:c43e: icmp_seq=56 ttl=64 time=0.843 ms

```

3. Temporary IPv6 Address

Enable Temporary IPv6 Address

```
sudo sysctl -w net.ipv6.conf.all.use_tempaddr=
```

Check Addresses

```
ip -6 addr show
```

```

kali@kali:~$ sudo sysctl -w net.ipv6.conf.all.use_tempaddr=2
net.ipv6.conf.all.use_tempaddr = 2

kali@kali:~$ ip -6 addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP qlen 1000
    inet6 2401:4900:8fed:de38:e61f:b06e:aba3:c43e/64 scope global dynamic noprefixroute
        valid_lft 81725sec preferred_lft 81725sec
    inet6 fe80::c677:7fec:f0d2:dffc/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

kali@kali:~$

```

4. Restaurant Search Script (Python)

```
restaurants = {  
    "Pizza Hut": "fd00:1::10",  
    "Dominos": "fd00:1::20",  
    "Burger King": "fd00:1::30"  
}  
  
search = input("Enter restaurant name: ").lower()  
  
for name, ipv6 in restaurants.items():  
    if search in name.lower():  
        print(name, "->", ipv6)
```

Example Input

pizza

Output

Pizza Hut -> fd00:1::10

5. AI-Generated IPv6 Subnetting Scenario

Configuration Command

```
sudo ip -6 addr add 2001:db8:1000:1::50/64 dev eth0
```

Verification

```
ip -6 addr show eth0
```